

Handout - *Loux & Crisp, “Time”*

I. The Problem: Is Time Real?

The contemporary metaphysics of time begins with a radical challenge from J.M.E. McTaggart who claimed that:

Time is unreal.

This is not skepticism about clocks or succession. It is the claim that the very structure we presuppose in thinking about change and temporal passage involves contradiction.

Why This Matters

If time is unreal:

- Change is illusory.
- Persistence through time is incoherent.
- Coming into and going out of existence are impossible.
- Many of our prephilosophical convictions are false.

Thus the challenge we face is the following:

Can we give a coherent metaphysical account of time that survives McTaggart's argument?

II. Background: Time and Change in the Tradition

Before McTaggart:

- **Aristotle:** Time is “the number of motion with respect to before and after.” (Physics IV, 11) Time presupposes change.
- Traditional metaphysics assumes:

- Objects persist.
- Properties change.
- The present differs from the past and future.

McTaggart radicalizes a traditional assumption:

If there is no change, there is no time.

His strategy:

1. Show that change requires the A-series.
2. Show that the A-series is contradictory.
3. Conclude that time is unreal.

III. McTaggart's Distinction: The A-Series and the B-Series

McTaggart argues that events and times can be ordered in two ways :

A. The B-Series

Ordered by:

- Earlier than
- Later than
- Simultaneous with

Features:

- Static.
- Each event has a fixed position.
- Relations never change.
- Fully describable in tenseless language.

Example:

The French Revolution is earlier than World War II.

This relation is *eternally* true.

B. The A-Series

Ordered by:

- Past
- Present
- Future

Features:

- Dynamic.
- Events change their A-properties.
- Involves **temporal becoming**.

An event:

- Is first future,
- Then present,
- Then past.

This captures the intuition of temporal passage.

IV. Part I of McTaggart's Argument

The B-Series Presupposes the A-Series

Step 1: Time Requires Change

McTaggart assumes (following tradition):

No change → No time.

Step 2: There Is No Change in the B-Series Alone

In the B-series:

- Events cannot come into existence.
- Events cannot cease to exist.

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- Events cannot gain or lose properties.
 - All B-facts are eternally true.

Example:

If an event is the death of Queen Anne, it is always that event .

Nothing about it changes.

Step 3: Only A-Properties Allow Change

The only possible change is:

- Change from future → present → past.

Thus:

The B-series is temporal only because of the A-series.

Conclusion of Part I:

The A-series is metaphysically fundamental.

V. Part II of McTaggart's Argument

The A-Series Is Contradictory

Key claim:

- Past, present, and future are mutually exclusive.
- Yet every event must be all three.

Objection:

They are had successively, not simultaneously.

McTaggart's reply:

To analyze "was" and "will be" requires:

- Appealing to further A-properties.
- Which generates a vicious infinite regress.

Conclusion:

- The A-series is impossible.
 - The B-series depends on the A-series.
 - Therefore, time is unreal .
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VI. The B-Theory (Eternalism)

B-theorists reject **Part I** of McTaggart's argument.

Core Thesis

Time is:

- A fourth dimension.
- A static spacetime manifold.
- Fully describable in tenseless terms.

All times are equally real.

This view is called:

Eternalism

Change on the B-Theory

Change = variation across time.

Distinction:

- Time-indexed property: warm-at- t_1
- Simpliciter property: warm

An object changes iff:

- It is F at t_1

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- Not-F at t_2

No contradiction.

No temporal passage required.

Objects are **four-dimensional spacetime worms**.

Scientific Support

Relativity theory:

- Denies an absolute present.
- Treats time as another dimension.
- Supports a block universe.

Thus B-theory aligns well with modern physics .

VII. The A-Theory

A-theorists reject **Part II** of McTaggart's argument.

Core Thesis

Time is irreducibly tensed.

The present is:

- Metaphysically privileged.
- Ontologically distinct.

Tensed language tracks real features of the world .

Objection to B-Theoretic Reductions

A-theorists argue:

- Tensed sentences cannot be reduced without loss.
 - Prior's example: "Thank goodness that's over!"
 - Relief presupposes real passage.
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VIII. Varieties of A-Theory

1. Spotlight Theory

- All times exist.
 - The present "moves" across them.
 - Present is ontologically illuminated.
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2. Growing Block Theory (Broad)

- Past and present are real.
 - Future is not.
 - Reality grows as new slices are added .
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3. Shrinking Block Theory

- Present and future are real.
 - Past no longer exists.
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4. Presentism

- Only the present exists .
 - Past and future entities do not exist.
 - Tensed propositions change truth value.
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IX. The New B-Theory

Developed in the 1980s .

Key shift:

- Abandon translation of tense.
- Provide **tenseless truth conditions** instead.

Tensed sentences are irreducible linguistically,
but their truthmakers are tenseless.

Smart (date-based truth conditions).

Mellor (token-reflexive truth conditions).

No tensed facts required.

X. The Deeper Fault Line

The debate turns on:

- Is change real?
- Is the present special?
- Is change mere variation or genuine becoming?
- Should metaphysics be constrained by physics?

The disagreement is not about vocabulary; it is about ontology.

XI. Comparative Table:

View	Response to Part I (B depends on A?)	Response to Part II (A is contradictory?)	Ontology of Time	Is Present Privileged?	Status of Change
Old B-Theory	Denies B depends on A	Often agrees A problematic	Eternalist block universe	No	Variation across time
New B-Theory	Same as above	Same	Eternalist; tenseless truthmakers	No	Variation; tenseless facts suffice
Spotlight A-Theory	Accepts need for A	Denies contradiction; passage primitive	All times exist	Yes (moving present)	Genuine becoming
Growing Block	Accepts need for A	Denies contradiction	Past + present real	Yes (leading edge)	Ontological growth
Shrinking Block	Accepts need for A	Denies contradiction	Present + future real	Yes (trailing edge)	Ontological loss
Presentism	Accepts need for A	Denies contradiction	Only present exists	Yes (only present real)	Truth-value change over time
McTaggart	B presupposes A	A contradictory → vicious regress	Time unreal	No time at all	Change impossible